Cloud Forensics Challenges in Bangladesh: A PRISMA Systematic Review with AI-Enhanced Resolution Strategies

Md. Ahsan Kabir | Dept. of CSE, United International University, Dhaka | mkabir2520016@mscse.uiu.ac.bd

LATEST REGULATORY UPDATE (2026)

The Cyber Security Ordinance 2025 REPEALED the Cyber Security Act 2023 outright -- the same cloud-evidence gap persists under the new law. NEW: the Personal Data Protection (Amendment) Ordinance 2026 (signed 5 Feb 2026) now requires a real-time in-country cloud copy for "restricted" data or Critical Information Infrastructure (CII) -- an incidental opening for future forensic-audit-access legislation that this review's framework is positioned to use.

1. Introduction & Motivation

Bangladesh: 100M+ internet-enabled devices; USD 1.2B+ public-sector cloud procurement since 2020. Investigative laws (ICT Act 2006, Cyber Security Ordinance 2025) were drafted for physical devices and are silent on cloud-hosted evidence. Result: a disproportionately high acquittal rate on cloud-evidence cases -- driven by chain-of-custody failures and Section 65B admissibility gaps, not disputed facts.

2. Research Gap & Questions

No prior PRISMA-based study maps cloud forensics challenges to Bangladesh's legal context.
RQ1: Technical cloud-forensics challenges in Bangladesh?
RQ2: Legal barriers to cloud-evidence admissibility?
RQ3: How can AI tools address RQ1/RQ2?
RQ4: What implementable framework improves Bangladesh's readiness?

3. Methodology (PRISMA 2020)

782 records identified -> 534 screened -> 48 full-text reviewed -> 16 studies included (2021-2026, open-access). 7 repositories, 6 query strings, single-reviewer thematic coding.

PRISMA 2020 Literature Selection Flow -- Cloud Forensics Challenges in Bangladesh

Records identified via 7-repository database search (arXiv, MDPI, Google Scholar, SSRN, NDC e-Journal, BJMIS, IJRIAS) n = 782	Duplicates removed n = 248
Records screened by title & abstract n = 534	Excluded as outside scope n = 486
Full-text articles assessed for eligibility n = 48	Excluded: paywalled n=18 tangential n=14
Studies included in final synthesis n = 16 (open-access, 2021-2026, English)	

4. Discussion & Contribution

First Bangladesh-specific PRISMA synthesis linking cloud forensics gaps to validated AI tools. Reframes the problem: the tools already exist -- the real gap is legal recognition and adoption, not missing technology. "Power of Disposal" (Karagiannis & Vergidis, 2021): only the cloud provider, not the Bangladeshi state, can currently grant access to cloud-hosted data.

5. Results: AI-CFRF-BD Framework

Five challenge clusters (C1-C5), each mapped to a validated AI/automation tool with quantitative evidence.

The AI-CFRF-BD Framework: Challenge Clusters Mapped to AI Resolution Tools

Cluster	Challenge	AI Resolution Tool	Metric
C1	Volatile Data & Log Expiry	CIAF -- Alharthi & Garcia, 2025 [5]	93% F1
C2	Multi-Tenancy & Evidence Contamination	ML-CF -- Tageldin & Venter, 2023 [7]	ISO/IEC 27043
C3	Chain-of-Custody & s.65B Admissibility	Blockchain CoC -- Igonor et al., 2025 [6]	14/17 jurisdictions
C4	Cross-Border MLAT Latency	CLERMS -- Akilal & Kechadi, 2025 [9]	40-70% MLAT reduction
C5	Forensic Readiness Deficit	Self-Hosted CF -- arXiv 2510.21246 [4]	60% cost reduction

Phase 1 (no legislative change): CIAF log hash-stamping (C1, 93% F1); ML-CF artefact isolation (C2); Self-Hosted CF pre-incident capture (C5).
Phase 2 (Ministry of Law practice note): Blockchain chain-of-custody (C3, compatible in 14/17 jurisdictions).
Phase 3 (inter-ministerial): CLERMS MLAT pre-validation (C4, 40-70% latency reduction).

6. Limitations & Future Work

Single-reviewer coding; English/open-access sources only. Framework is synthesis-level -- not yet piloted on Bangladeshi data. Next steps: NTMC pilot of a Phase-1 tool; a narrow forensic-audit-access amendment leveraging the new CII residency rule; practitioner consultation with CID Digital Forensics & BGD e-GOV CIRT.

References (selected)

[1] Alshabibi et al. 2024 · [8] Karagiannis & Vergidis 2021 · [11] Uzzal 2025 · [15] Sharmin & Rahman 2025 · [20] Cyber Security Ordinance 2025 · [21] PDPO (Amendment) Ordinance 2026
Full 22-reference list in the accompanying paper.